HAJRAH SALEHA KAZI

AI/ML Engineer | Applied AI | Computer Vision | Software Engineering

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PROFESSIONAL SUMMARY

AI/ML-focused Computer Science student and technical team lead with hands-on experience in computer vision, multimodal ML, video intelligence and full-stack development. Led teams delivering AI solutions for industrial quality inspection, livestock monitoring, video summarization and productivity analytics. Published AI researcher; institute rank holder.

EDUCATION

B.Tech – Computer Science and Engineering (AI & ML) Aug 2026 – May 2029
Vishwakarma Institute of Technology (VIT), Pune | Direct Second-Year Entry

Diploma – Computer Engineering Jul 2023 – Jul 2026
Trinity Polytechnic, Pune | First Class with Distinction | Final Semester: 96.12% | College Rank 1 (six consecutive semesters)

TECHNICAL SKILLS

Core AI/ML: Python, Machine Learning, Deep Learning, Computer Vision, PyTorch, TensorFlow, YOLO, OpenCV, Natural Language Processing (NLP), Generative AI, Hugging Face, FastAPI
AI/ML Tools: YOLOv11, YOLO26, Scikit-learn, NumPy, Pandas, VADER, MFCC, Multimodal AI, LLM Integration, Sentiment Analysis, Model Training & Evaluation
Languages, Backend & Tools: C++, Java, REST APIs, WebSockets, Next.js 15, React, HTML5, CSS3, Git, GitHub, Google Colab, Vercel
Concepts: Data Structures and Algorithms, Object-Oriented Programming, Real-Time Inference, API Integration, Data Analysis

EXPERIENCE

AI/ML Lead Developer – Industry-Sponsored Project | **MESAKI** Sep 2026 – Present
Right-Fit AI-CV: Real-Time Computer Vision for Adhesive Quality Inspection

- Lead a 6-member team developing a YOLO-based computer vision system for automated footwear quality inspection across surface-finish and adhesive-application defects (residue, finishing imperfections, insufficient coverage, gaps, overflow, uneven application).
- Execute image annotation and model training in Google Colab across **8 defect categories** and **2,500 annotated images**, achieving **90% mAP@50**, **88% precision**, **86% recall** and inference latency below **100 ms**.

AI Research Intern / Project Lead | **MESAKI** Feb 2026 – Apr 2026
GoatAI: Real-Time Livestock Activity Monitoring

- Led a 3-member team developing a computer vision system that analyzes CCTV footage to identify goat activity across **4 classes** (sleeping, drinking, fighting, general) and generate automated summaries for farmers.
- Processed **20 hours** of CCTV footage, achieving **85% activity-classification accuracy** at **15 FPS**, with support for up to **10 goats per frame**.

AI Intern / Lead Developer | **Cognizeal (CogniLABs)** Jun 2025 – Dec 2025
RecapBot: AI-Powered Video Analysis and Summarization

- Led a 2-member team building a video-understanding application supporting videos up to **30 minutes**, processing a 10-minute video in under **2 minutes** through a **3-stage pipeline** (video processing, content understanding, summarization); contributed to video analysis and backend integration.

Data Science & Analytics Intern | **Rajavi Technologies Pvt. Ltd.** Jun 2025 – Aug 2025
FocusPulse Elite: AI-Powered Productivity Analytics

- Independently developed a productivity analytics platform with **8 productivity indicators**, **5 analytics dashboard modules** and local-first monitoring, validated across **100 test sessions** at **95% tracking reliability**.
- Implemented START/STOP session controls, visual dashboards, AI-powered insights and personalized recommendations.

SELECTED PROJECTS & RESEARCH

NeuroCalm – Multimodal Affective Computing Framework for Mental Health Monitoring Published Research

- Developed a full-stack AI platform integrating facial-expression analysis, acoustic emotion recognition and context-aware NLP across **3 modalities**.
- Fine-tuned YOLOv11 on the MEFC dataset to recognize **7 emotional states** with **5-frame temporal smoothing**; built an acoustic pipeline on an **18-dimensional feature vector** (MFCCs, pitch, jitter, shimmer, spectral bandwidth) with an MLP classifier.
- Integrated VADER sentiment analysis and Google Gemini 2.x; implemented a FastAPI backend with WebSockets and a Next.js 15 frontend with private journaling, mood tracking, guided breathing and analytics.
- Achieved **15–25 FPS** CPU-only visual inference and audio latency below **250 ms**. Published in the *Int. Journal of Innovative Research in Technology*.

LEADERSHIP & ADDITIONAL EXPERIENCE

- **Founder & Creative Director, Visionary Designers Co.** | Aug 2025 – Present – Lead creative direction and client delivery for branding and design.
- **Technical Team Leader, Trinity Polytechnic Coding Club** | Aug 2025 – May 2026 – Built certificate-generation module; supported website and Vercel deployment; organized hackathons.
- **Udemy Instructor** | Sep 2023 – Present – Create and deliver programming and technology courses.
- **IGCSE ICT & Secondary Teacher, SCUBE Learning Center** – Deliver ICT instruction and build students' practical computing skills.
- **Technical & Digital Operations Volunteer, TIQ** | Nov 2020 – Present – Support English, Urdu and Marathi departments: quiz publishing, results, certificates, Zoom, YouTube Live.

CERTIFICATIONS

GenAI-Powered Data Analytics Job Simulation – Tata Group iQ (Jul 2026) | AWS AI Practitioner Challenge (Apr 2026) | Artificial Intelligence Fundamentals – IBM (Mar 2026) | YUVA AI For All – TCS iON (Feb 2026) | Exploring Networking with Cisco Packet Tracer – Cisco Networking Academy (Sep 2026)

ACHIEVEMENTS & PUBLICATIONS

- **Best Student Award, Trinity Polytechnic:** topper across departments; **College Rank 1**, six consecutive semesters.
- **Academic Excellence Awards:** 2024–25, 2025–26. **First Prize, Smart India Hackathon 2025 Internal Hackathon:** cryptocurrency address collection and categorization.
- **Published Researcher & Author:** NeuroCalm paper; authored *Pieces of Me, Still Whole* (first student-author from the college to publish a book).

LANGUAGES

English | Hindi | Marathi | Urdu | Arabic